1.10 Series: Method of Differences

Worked Solutions

1 a)

$$\begin{array}{ll} 1 & -\cancel{\frac{1}{2}} \\ +\cancel{\frac{1}{2}} & -\cancel{\frac{1}{3}} \\ +\cancel{\frac{1}{3}} & -\cancel{\frac{1}{4}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{23}} & -\cancel{\frac{1}{24}} \\ +\cancel{\frac{1}{24}} & -\frac{1}{25} \end{array}$$

$$= 1 - \frac{1}{25} = \frac{24}{25}$$

1 b)

$$\begin{array}{ll} 1 & -\cancel{\frac{1}{4}} \\ +\cancel{\frac{1}{4}} & -\cancel{\frac{1}{7}} \\ +\cancel{\frac{1}{7}} & -\cancel{\frac{1}{10}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{3n-5}} & -\cancel{\frac{1}{3n-2}} \\ +\cancel{\frac{1}{3n-2}} & -\frac{1}{3n+1} \end{array}$$

$$= 1 - \frac{1}{3n+1}$$

1 c)

$$\begin{array}{ll} \frac{1}{6} & -\cancel{\frac{1}{10}} \\ +\frac{1}{8} & -\cancel{\frac{1}{12}} \\ +\cancel{\frac{1}{10}} & -\cancel{\frac{1}{14}} \\ +\cancel{\frac{1}{12}} & -\cancel{\frac{1}{16}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{2n}} & -\cancel{\frac{1}{2(n+2)}} \\ +\cancel{\frac{1}{2(n+1)}} & -\frac{1}{2(n+3)} \\ +\cancel{\frac{1}{2(n+2)}} & -\frac{1}{2(n+4)} \end{array}$$

$$= \frac{1}{6} + \frac{1}{8} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)} = \frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)}$$

1 d)

$$\begin{array}{rr} \cancel{2} & -1 \\ \cancel{+3} & \cancel{-2} \\ \cancel{+4} & \cancel{-3} \\ \cancel{\cdot\cdot} & \cancel{\cdot\cdot} \\ \cancel{+(n-1)} & \cancel{-(n-2)} \\ \cancel{+n} & \cancel{-(n-1)} \\ +(n+1) & \cancel{-n} \end{array}$$

$$= -1 + (n+1) = n$$

1 e)

$$\begin{array}{rr} \cancel{1\times 2} & -1\times 0 \\ \cancel{+2\times 3} & \cancel{-2\times 1} \\ \cancel{+3\times 4} & \cancel{-3\times 2} \\ \cancel{+4\times 5} & \cancel{-4\times 3} \\ \cancel{\cdot\cdot} & \cancel{\cdot\cdot} \\ \cancel{+(n-1)n} & \cancel{-(n-1)(n-2)} \\ +n(n+1) & \cancel{-n(n-1)} \end{array}$$

$$= -0 + n(n+1) = n(n+1)$$

2 a)

$$\frac{1}{n+1} - \frac{1}{n+2} = \frac{n+2}{(n+1)(n+2)} - \frac{n+1}{(n+1)(n+2)}$$
$$= \frac{n+2-n-1}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

2 b)

$$1 + \frac{1}{n+1} = \frac{n+1}{n+1} + \frac{1}{n+1} = \frac{n+2}{n+1}$$

2 c)

$$\frac{1}{2} - \frac{1}{n+2} = \frac{n+2}{2(n+2)} - \frac{2}{2(n+2)} = \frac{n+2-2}{2(n+2)} = \frac{n}{2(n+2)}$$

2 d)

$$\begin{aligned}
&-\frac{3}{2} + \frac{1}{n+2} + \frac{1}{n+1} \\
&= -\frac{3(n+1)(n+2)}{2(n+1)(n+2)} + \frac{2(n+1)}{2(n+1)(n+2)} + \frac{2(n+2)}{2(n+1)(n+2)} \\
&= -\frac{3n^2+9n+2}{2(n+1)(n+2)} + \frac{2n+2}{2(n+1)(n+2)} + \frac{2n+4}{2(n+1)(n+2)} \\
&= \frac{-3n^2-9n-2+2n+2+2n+4}{2(n+1)(n+2)} \\
&= \frac{-3n^2-5n}{2(n+1)(n+2)} \\
&= \frac{-n(3n+5)}{2(n+1)(n+2)}
\end{aligned}$$

2 e)

$$\begin{aligned}
&\frac{1}{12} + \frac{1}{2(n+3)} - \frac{1}{2(n+2)} \\
&= \frac{(n+2)(n+3)}{12(n+2)(n+3)} + \frac{6(n+2)}{12(n+2)(n+3)} - \frac{6(n+3)}{12(n+2)(n+3)} \\
&= \frac{n^2+5n+6}{12(n+2)(n+3)} + \frac{6n+12}{12(n+2)(n+3)} - \frac{6n+18}{12(n+2)(n+3)} \\
&= \frac{n^2+5n+6+6n+12-6n-18}{12(n+2)(n+3)} \\
&= \frac{n^2+5n}{12(n+2)(n+3)}
\end{aligned}$$

3)

$$\sum_{r=1}^{50} \left(\frac{1}{r+2} - \frac{1}{r} \right) =$$

$$
\begin{array}{cc}
\cancel{\frac{1}{3}} & -1 \\
\cancel{+\frac{1}{4}} & -\frac{1}{2} \\
\cancel{+\frac{1}{5}} & \cancel{-\frac{1}{3}} \\
\cancel{+\frac{1}{6}} & \cancel{-\frac{1}{4}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{50}} & \cancel{-\frac{1}{48}} \\
+\frac{1}{51} & \cancel{-\frac{1}{49}} \\
+\frac{1}{52} & \cancel{-\frac{1}{50}}
\end{array}
$$

$$= -1 - \frac{1}{2} + \frac{1}{51} + \frac{1}{52} = -\frac{3875}{2652}$$

4)

$$\sum_{r=8}^{100} \left(\frac{1}{r+1} - \frac{1}{r} \right) =$$

$$
\begin{array}{cc}
\cancel{\frac{1}{9}} & -\frac{1}{8} \\
\cancel{+\frac{1}{10}} & \cancel{-\frac{1}{9}} \\
\cancel{+\frac{1}{11}} & \cancel{-\frac{1}{10}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{99}} & \cancel{-\frac{1}{98}} \\
\cancel{+\frac{1}{100}} & \cancel{-\frac{1}{99}} \\
+\frac{1}{101} & \cancel{-\frac{1}{100}}
\end{array}
$$

$$= -\frac{1}{8} + \frac{1}{101} = -\frac{93}{808}$$

5)

$$\sum_{r=1}^{n} \left(\frac{1}{r+8} - \frac{1}{r+7} \right) =$$

$$\begin{array}{ll} \cancel{\frac{1}{9}} & -\frac{1}{8} \\ +\cancel{\frac{1}{10}} & -\cancel{\frac{1}{9}} \\ +\cancel{\frac{1}{11}} & -\cancel{\frac{1}{10}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{n+6}} & -\cancel{\frac{1}{n+5}} \\ +\cancel{\frac{1}{n+7}} & -\cancel{\frac{1}{n+6}} \\ +\frac{1}{n+8} & -\cancel{\frac{1}{n+7}} \end{array}$$

$$= -\frac{1}{8} + \frac{1}{n+8}$$

6)

$$\sum_{r=1}^{n} \left(\frac{1}{r+3} - \frac{1}{r+4} \right) =$$

$$\begin{array}{ll} \frac{1}{4} & -\cancel{\frac{1}{5}} \\ +\cancel{\frac{1}{5}} & -\cancel{\frac{1}{6}} \\ +\cancel{\frac{1}{6}} & -\cancel{\frac{1}{7}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{n+1}} & -\cancel{\frac{1}{n+2}} \\ +\cancel{\frac{1}{n+2}} & -\cancel{\frac{1}{n+3}} \\ +\cancel{\frac{1}{n+3}} & -\frac{1}{n+4} \end{array}$$

$$= \frac{1}{4} - \frac{1}{n+4}$$

$$= \frac{n+4}{4(n+4)} - \frac{4}{4(n+4)} = \frac{n}{4(n+4)} \text{ as req.}$$

7)

$$\sum_{r=1}^{n} \left(\frac{1}{r+5} - \frac{1}{r+3} \right) =$$

$$
\begin{array}{ll}
\cancel{\frac{1}{6}} & -\frac{1}{4} \\
\cancel{+\frac{1}{7}} & -\frac{1}{5} \\
\cancel{+\frac{1}{8}} & \cancel{-\frac{1}{6}} \\
\cancel{+\frac{1}{9}} & \cancel{-\frac{1}{7}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{n+3}} & \cancel{-\frac{1}{n+1}} \\
+\frac{1}{n+4} & \cancel{-\frac{1}{n+2}} \\
+\frac{1}{n+5} & \cancel{-\frac{1}{n+3}}
\end{array}
$$

$$
\begin{aligned}
&= -\frac{1}{4} - \frac{1}{5} + \frac{1}{n+4} + \frac{1}{n+5} \\
&= -\frac{9}{20} + \frac{1}{n+4} + \frac{1}{n+5}
\end{aligned}
$$

8 a)

$$
\begin{aligned}
\frac{1}{r+4} - \frac{1}{r+5} &= \frac{r+5}{(r+4)(r+5)} - \frac{r+4}{(r+4)(r+5)} \\
&= \frac{r+5-r-4}{(r+4)(r+5)} \\
&= \frac{1}{(r+4)(r+5)} \text{ as req.}
\end{aligned}
$$

8 b)

$$
\sum_{r=1}^{n} \frac{1}{(r+4)(r+5)} = \sum_{r=1}^{n} \left(\frac{1}{r+4} - \frac{1}{r+5} \right) =
$$

$$
\begin{array}{ll}
\frac{1}{5} & \cancel{-\frac{1}{6}} \\
\cancel{+\frac{1}{6}} & \cancel{-\frac{1}{7}} \\
\cancel{+\frac{1}{7}} & \cancel{-\frac{1}{8}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{n+2}} & \cancel{-\frac{1}{n+3}} \\
\cancel{+\frac{1}{n+3}} & \cancel{-\frac{1}{n+4}} \\
\cancel{+\frac{1}{n+4}} & -\frac{1}{n+5}
\end{array}
$$

$$
= \frac{1}{5} - \frac{1}{n+5}
$$

9 a)

$$\begin{aligned}\frac{1}{r+1}-\frac{1}{r+2} &= \frac{r+2}{(r+1)(r+2)}-\frac{r+1}{(r+1)(r+2)}\\ &= \frac{r+2-r-1}{(r+1)(r+2)}\\ &= \frac{1}{(r+1)(r+2)}\end{aligned}$$

9 b)

$$\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}=\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+2}\right)=$$

$$\begin{array}{cc}\frac{1}{2} & \cancel{-\frac{1}{3}}\\ \cancel{+\frac{1}{3}} & \cancel{-\frac{1}{4}}\\ \cancel{+\frac{1}{4}} & \cancel{-\frac{1}{5}}\\ \cancel{\cdots} & \cancel{\cdots}\\ \cancel{+\frac{1}{n-1}} & \cancel{-\frac{1}{n}}\\ \cancel{+\frac{1}{n}} & \cancel{-\frac{1}{n+1}}\\ \cancel{+\frac{1}{n+1}} & -\frac{1}{n+2}\end{array}$$

$$\begin{aligned}&= \frac{1}{2}-\frac{1}{n+2}\\ &= \frac{n+2}{2(n+2)}-\frac{2}{2(n+2)}=\frac{n}{2n+4}\end{aligned}$$

10 a)

$$\begin{aligned}\frac{1}{r-1}-\frac{1}{r+1} &= \frac{r+1}{(r-1)(r+1)}-\frac{r-1}{(r-1)(r+1)}\\ &= \frac{r+1-r+1}{(r-1)(r+1)}\\ &= \frac{2}{(r-1)(r+1)} \text{ as req.}\end{aligned}$$

10 b)

$$\frac{1}{r^2-1}=\frac{1}{(r-1)(r+1)}$$

$$\sum_{r=5}^{50}\frac{1}{r^2-1}=\frac{1}{2}\sum_{r=5}^{50}\frac{2}{(r-1)(r+1)}=\frac{1}{2}\sum_{r=5}^{50}\left(\frac{1}{r-1}-\frac{1}{r+1}\right)$$

Since

$$\sum_{r=5}^{50}\left(\frac{1}{r-1}-\frac{1}{r+1}\right)=$$

$$\begin{array}{ll} \frac{1}{4} & -\cancel{\frac{1}{6}} \\ +\frac{1}{5} & -\cancel{\frac{1}{7}} \\ +\cancel{\frac{1}{6}} & -\cancel{\frac{1}{8}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{47}} & -\cancel{\frac{1}{49}} \\ +\cancel{\frac{1}{48}} & -\frac{1}{50} \\ +\cancel{\frac{1}{49}} & -\frac{1}{51} \end{array}$$

$$= \frac{1}{4} + \frac{1}{5} - \frac{1}{50} - \frac{1}{51}$$

$$= \frac{2093}{5100}$$

$$\frac{1}{2}\sum_{r=5}^{50}\left(\frac{1}{r-1} - \frac{1}{r+1}\right) = \frac{2093}{10200}$$

11 a)

$$\frac{1}{r+3} - \frac{1}{r+4} = \frac{r+4}{(r+3)(r+4)} - \frac{r+3}{(r+3)(r+4)}$$

$$= \frac{1}{(r+3)(r+4)} = \frac{1}{r^2+7r+12} \text{ as req.}$$

11 b)

$$\sum_{r=8}^{22}\frac{1}{r^2+7r+12} = \sum_{r=8}^{22}\left(\frac{1}{r+3} - \frac{1}{r+4}\right) =$$

$$\begin{array}{ll} \frac{1}{11} & -\cancel{\frac{1}{12}} \\ +\cancel{\frac{1}{12}} & -\cancel{\frac{1}{13}} \\ +\cancel{\frac{1}{33}} & -\cancel{\frac{1}{14}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{23}} & -\cancel{\frac{1}{24}} \\ +\cancel{\frac{1}{24}} & -\cancel{\frac{1}{25}} \\ +\cancel{\frac{1}{25}} & -\frac{1}{26} \end{array}$$

$$= \frac{1}{11} - \frac{1}{26} = \frac{15}{286}$$

12 a)

$$\begin{aligned}\frac{1}{r-1}-\frac{1}{r+2} &= \frac{r+2}{(r-1)(r+2)}-\frac{r-1}{(r-1)(r+2)}\\ &= \frac{2--1}{(r-1)(r+2)}\\ &= \frac{3}{(r-1)(r+2)} \text{ as req.}\end{aligned}$$

12 b)

$$\sum_{r=2}^{n}\frac{1}{(r-1)(r+2)} = \frac{1}{3}\sum_{r=2}^{1}\frac{3}{(r-1)(r+2)} = \frac{1}{3}\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right)$$

Since

$$\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right) =$$

$$\begin{array}{rr} 1 & \cancel{-\frac{1}{4}} \\ +\frac{1}{2} & \cancel{-\frac{1}{5}} \\ +\frac{1}{3} & \cancel{-\frac{1}{6}} \\ \cancel{+\frac{1}{4}} & \cancel{-\frac{1}{7}} \\ \cancel{\cdots} & \cancel{\cdots} \\ \cancel{+\frac{1}{n-3}} & -\frac{1}{n} \\ \cancel{+\frac{1}{n-2}} & -\frac{1}{n+1} \\ \cancel{+\frac{1}{n-1}} & -\frac{1}{n+2} \end{array}$$

$$\begin{aligned}&= 1+\frac{1}{2}+\frac{1}{3}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}\\ &= \frac{11}{6}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}\end{aligned}$$

$$\frac{1}{3}\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right) = \frac{1}{3}\left(\frac{11}{6}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}\right)$$

13)

$$\begin{aligned}\frac{1}{r^3}-\frac{1}{(r+1)^3} &= \frac{(r+1)^3}{r^3(r+1)^3}-\frac{r^3}{r^3(r+1)^3}\\ &= \frac{(r+1)^3-r^3}{r^3(r+1)^3}\\ &= \frac{r^3+3r^2+3r+1-r^3}{r^3(r+1)^3}\\ &= \frac{3r^2+3r+1}{r^3(r+1)^3}\end{aligned}$$

$$\therefore \sum_{r=1}^{n} \frac{3r^2+3r+1}{r^3(r+1)^3} = \sum_{r=1}^{n} \left(\frac{1}{r^3} - \frac{1}{(r+1)^3} \right)$$

$$\begin{array}{rr} 1 & -\cancel{\frac{1}{2^3}} \\ +\cancel{\frac{1}{2^3}} & -\cancel{\frac{1}{3^3}} \\ +\cancel{\frac{1}{3^3}} & -\cancel{\frac{1}{4^3}} \\ +\cancel{\frac{1}{4^3}} & -\cancel{\frac{1}{5^3}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{(n-2)^3}} & -\cancel{\frac{1}{(n-1)^2}} \\ +\cancel{\frac{1}{(n-1)^3}} & -\cancel{\frac{1}{n^3}} \\ +\cancel{\frac{1}{n^3}} & -\frac{1}{(n+1)^3} \end{array}$$

$$= 1 - \frac{1}{(n+1)^3}$$

14)

$$\frac{1}{r-2} - \frac{1}{r} = \frac{r}{r(r-2)} - \frac{r-2}{r(r-2)} = \frac{2}{r(r-2)}$$

$$\therefore \sum_{r=3}^{n} \frac{1}{r^2-2r} = \frac{1}{2} \sum_{r=3}^{n} \frac{2}{r(r-2)} = \frac{1}{2} \sum_{r=3}^{n} \left(\frac{1}{r-2} - \frac{1}{r} \right)$$

Since

$$\sum_{r=3}^{n} \left(\frac{1}{r-2} - \frac{1}{r} \right) =$$

$$\begin{array}{rr} 1 & -\cancel{\frac{1}{3}} \\ +\frac{1}{2} & -\cancel{\frac{1}{4}} \\ +\cancel{\frac{1}{3}} & -\cancel{\frac{1}{5}} \\ +\cancel{\frac{1}{4}} & -\cancel{\frac{1}{6}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{n-4}} & -\cancel{\frac{1}{n-2}} \\ +\cancel{\frac{1}{n-3}} & -\frac{1}{n-1} \\ +\cancel{\frac{1}{n-2}} & -\frac{1}{n} \end{array}$$

$$= 1 + \frac{1}{2} - \frac{1}{n-1} - \frac{1}{n}$$

$$= \frac{3}{2} - \frac{1}{n-1} - \frac{1}{n}$$

$$\sum_{r=3}^{n} \frac{1}{r^2-2r} = \frac{1}{2}\left(\frac{3}{2} - \frac{1}{n-1} - \frac{1}{n}\right)$$
$$= \frac{1}{2}\left(\frac{3n(n-1)}{2n(n-1)} - \frac{2n}{2n(n-1)} - \frac{2(n-1)}{2n(n-1)}\right)$$
$$= \frac{1}{2}\left(\frac{3n^2-3n}{2n(n-1)} - \frac{2n}{2n(n-1)} - \frac{2n-2}{2n(n-1)}\right)$$
$$= \frac{1}{2}\left(\frac{3n^2-7n+2}{2n(n-1)}\right)$$
$$= \frac{(3n-1)(n-2)}{4n(n-1)}$$

15 a)

$$\frac{1}{2r+1} - \frac{1}{2r+3} = \frac{2r+3}{(2r+1)(2r+3)} - \frac{2r+1}{(2r+1)(2r+3)}$$
$$= \frac{2}{(2r+1)(2r+3)}$$

15 b)

$$\sum_{r=1}^{n} \frac{2}{(2r+1)(2r+3)} = \sum_{r=1}^{n}\left(\frac{1}{2r+1} - \frac{1}{2r+3}\right) =$$

$$\begin{array}{cc}
\frac{1}{3} & \cancel{-\frac{1}{5}} \\
\cancel{+\frac{1}{5}} & \cancel{-\frac{1}{7}} \\
\cancel{+\frac{1}{7}} & \cancel{-\frac{1}{9}} \\
\cancel{+\frac{1}{9}} & \cancel{-\frac{1}{11}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{2n-3}} & \cancel{-\frac{1}{2n-1}} \\
\cancel{+\frac{1}{2n-1}} & \cancel{-\frac{1}{2n+1}} \\
\cancel{+\frac{1}{2n+1}} & -\frac{1}{2n+3}
\end{array}$$
$$= \frac{1}{3} - \frac{1}{2n+3}$$

So since there are 15 terms in the series, the sum is $\frac{1}{3} - \frac{1}{30+3} = \frac{10}{33}$

16 a)

$$\frac{1}{r} - \frac{1}{r+1} = \frac{r+1}{r(r+1)} - \frac{r}{r(r+1)} = \frac{1}{r(r+1)}$$

16 b)

$$\sum_{r=1}^{n} \frac{1}{r(r+1)} = \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1} \right) =$$

$$\begin{array}{ll} \frac{1}{1} & -\cancel{\frac{1}{2}} \\ +\cancel{\frac{1}{2}} & -\cancel{\frac{1}{3}} \\ +\cancel{\frac{1}{3}} & -\cancel{\frac{1}{4}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{n-2}} & -\cancel{\frac{1}{n-1}} \\ +\cancel{\frac{1}{n-1}} & -\cancel{\frac{1}{n}} \\ +\cancel{\frac{1}{n}} & -\frac{1}{n+1} \end{array}$$

$$= 1 - \frac{1}{n+1}$$

So $1 - \frac{1}{n+1} > 0.98$

$$n + 1 - 1 > 0.98(n+1)$$
$$n > 0.98n + 0.98$$
$$0.02n > 0.98$$
$$n > 49$$

So since it needs to exceed 0.98 then $n = 50$

So 50 terms

17 a)

$$\frac{1}{2r} - \frac{1}{2r+2} = \frac{2r+2}{2r(2r+2)} - \frac{2r}{2r(2r+2)}$$
$$= \frac{2}{2r(2r+2)} = \frac{1}{2r(r+1)}$$

17 b)

$$\sum_{r=1}^{n} \frac{1}{2r(r+1)} = \sum_{r=1}^{n} \left(\frac{1}{2r} - \frac{1}{2r+2} \right) =$$

$$\begin{array}{ll} \frac{1}{2} & -\cancel{\frac{1}{4}} \\ +\cancel{\frac{1}{4}} & -\cancel{\frac{1}{6}} \\ +\cancel{\frac{1}{6}} & -\cancel{\frac{1}{8}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{1}{n-2}} & -\cancel{\frac{1}{2n-2}} \\ +\cancel{\frac{1}{2n-2}} & -\cancel{\frac{1}{2n}} \\ +\cancel{\frac{1}{2n}} & -\frac{1}{2n+2} \end{array}$$

$$= \frac{1}{2} - \frac{1}{2n+2}$$

17 c)

As $n \to \infty$, $\frac{1}{2n+2} \to 0$

$$\sum_{r=1}^{\infty} \frac{1}{2r(r+1)} = \frac{1}{2} - 0 = \frac{1}{2}$$

18 a)

$$\begin{aligned}
&\frac{1}{2}[(r+1)(r+2) - r(r+1)] \\
&= \frac{1}{2}(r+1)[r+2-r] \\
&= \frac{1}{2}(r+1) \times 2 \\
&= (r+1) \text{ as req.}
\end{aligned}$$

18 b)

$$\sum_{r=1}^{n} 2(r+1) = \sum_{r=1}^{n} 2 \times \frac{1}{2}[(r+1)(r+2) - r(r+1)]$$

$$= \sum_{r=1}^{n} (r+1)(r+2) - r(r+1) =$$

$$\begin{array}{rr}
\cancel{2 \times 3} & -1 \times 2 \\
\cancel{+3 \times 4} & \cancel{-2 \times 3} \\
\cancel{+4 \times 5} & \cancel{-3 \times 4} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+(n-1)n} & \cancel{-(n-1)(n+1)} \\
\cancel{+n(n+1)} & \cancel{-(n-1)n} \\
+(n+1)(n+2) & \cancel{-n(n+1)}
\end{array}$$

$$\begin{aligned}
&= (n+1)(n+2) - 1 \times 2 \\
&= (n+1)(n+2) - 2 \\
&= n^2 + 3n
\end{aligned}$$

19 a)

$$\begin{aligned}
&\frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)] \\
&= \frac{1}{3}(r+1)(r+2)[r+3-r] \\
&= \frac{1}{3}(r+1)(r+2) \times 3 \\
&= (r+1)(r+2) \text{ as req.}
\end{aligned}$$

19 b)

$$\sum_{r=1}^{n} 3(r+1)(r+2) = \sum_{r=1}^{n} 3 \times \frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)]$$

$$= \sum_{r=1}^{n} (r+1)(r+2)(r+3) - r(r+1)(r+2) =$$

$$\begin{array}{rr}
\cancel{2 \times 3 \times 4} & -1 \times 2 \times 3 \\
\cancel{+3 \times 4 \times 5} & \cancel{-2 \times 3 \times 4} \\
\cancel{+4 \times 5 \times 6} & \cancel{-3 \times 4 \times 5} \\
\cancel{\ldots} & \cancel{\ldots} \\
\cancel{+(n-1)n(n+1)} & \cancel{-(n-1)(n+1)(n+2)} \\
\cancel{+n(n+1)(n+2)} & \cancel{-(n-1)n(n+1)} \\
+(n+1)(n+2)(n+3) & \cancel{-n(n+1)(n+2)}
\end{array}$$

$$= (n+1)(n+2)(n+3) - 1 \times 2 \times 3$$

$$= (n+1)(n+2)(n+3) - 6$$

20)

$$\sum_{r=1}^{n} \left(\frac{2}{r^2} - \frac{2}{(r+1)^2} \right) =$$

$$\begin{array}{rr}
\frac{2}{1} & \cancel{-\frac{2}{4}} \\
\cancel{+\frac{2}{4}} & \cancel{-\frac{2}{9}} \\
\cancel{+\frac{2}{9}} & \cancel{-\frac{2}{16}} \\
\cancel{\ldots} & \cancel{\ldots} \\
\cancel{+\frac{2}{(n-2)^2}} & \cancel{-\frac{2}{(n-1)^2}} \\
\cancel{+\frac{2}{(n-1)^2}} & \cancel{-\frac{2}{n^2}} \\
\cancel{+\frac{2}{n^2}} & -\frac{2}{(n+1)^2}
\end{array}$$

$$= 2 - \frac{2}{(n+1)^2}$$

So

$$2 - \frac{2}{(n+1)^2} > 1.95$$

$$0.05 > \frac{2}{(n+1)^2}$$

$$0.05(n+1)^2 > 2$$

$$0.05n^2 + 0.1n - 1.95 > 0$$

$$n > 5.32456 \ldots \therefore n = 6$$

21 a)

$$\frac{1}{r!} - \frac{1}{(r+1)!} = \frac{r+1}{(r+1)r!} - \frac{1}{(r+1)r!} = \frac{r}{(r+1)r!} \text{ as req.}$$

21 b)

$$\sum_{r=1}^{n} \frac{r}{(r+1)r!} = \sum_{r=1}^{n} \left(\frac{1}{r!} - \frac{1}{(r+1)!} \right) =$$

$$\begin{array}{ll}
\frac{1}{1!} & \cancel{-\frac{1}{2!}} \\
\cancel{+\frac{1}{2!}} & \cancel{-\frac{1}{3!}} \\
\cancel{+\frac{1}{3!}} & \cancel{-\frac{1}{4!}} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+\frac{1}{(n-2)!}} & \cancel{-\frac{1}{(n-1)!}} \\
\cancel{+\frac{1}{(n-1)!}} & \cancel{-\frac{1}{n!}} \\
\cancel{+\frac{1}{n!}} & -\frac{1}{(n+1)!}
\end{array}$$

$$= 1 - \frac{1}{(n+1)!}$$

22 a)

$$(r+2)^3 - r^3 = r^3 + 6r^2 + 12r + 8 - r^3 = 6r^2 + 12r + 8$$

22 b)

$$\sum_{r=1}^{n} 3r^2 + 6r + 4 = \frac{1}{2} \sum_{r=1}^{n} ((r+2)^3 - r^3)$$

Since

$$\sum_{r=1}^{n} ((r+2)^3 - r^3) =$$

$$\begin{array}{ll}
\cancel{3^3} & -1^3 \\
\cancel{+4^3} & -2^3 \\
\cancel{+5^3} & \cancel{-3^3} \\
\cancel{\cdots} & \cancel{\cdots} \\
\cancel{+n^3} & \cancel{-(n-2)^3} \\
+(n+1)^3 & \cancel{-(n-1)^3} \\
+(n+2)^3 & \cancel{-n^3}
\end{array}$$

$$= -1 - 8 + (n+1)^3 + (n+2)^3$$
$$= -9 + (n+1)^3 + (n+2)^3$$
$$\therefore \sum_{r=1}^{n} 3r^2 + 6r + 4 = -\frac{9}{2} + \frac{1}{2}(n+1)^3 + \frac{1}{2}(n+2)^3$$
$$= -\frac{9}{2} + \frac{1}{2}(n^3 + 3n^2 + 3n + 1) + \frac{1}{2}(n^3 + 6n^2 + 12n + 8)$$
$$= -\frac{9}{2} + \frac{1}{2}(n^3 + 3n^2 + 3n + 1) + \frac{1}{2}(n^3 + 6n^2 + 12n + 8)$$
$$= n^3 + \frac{9}{2}n^2 + \frac{15}{2}n$$
$$= \frac{1}{2}n(2n^2 + 9n + 15)$$

22 c)

$$\sum_{r=1}^{n} r(r+2) = \sum_{r=1}^{n} r^2 + 2r$$
$$= \frac{1}{3}\left(\sum_{r=1}^{n} 3r^2 + 6r + 4\right) - \frac{1}{3}\left(\sum_{r=1}^{n} 4\right)$$
$$= \frac{1}{3} \times \frac{1}{2}n(2n^2 + 9n + 15) - \frac{1}{3} \times 4n$$
$$= \frac{1}{6}n(2n^2 + 9n + 15 - 8)$$
$$= \frac{1}{6}n(2n^2 + 9n + 7)$$
$$= \frac{1}{6}n(n+1)(2n+7)$$

23 a)

$$(2r+1)^3 - (2r-1)^3 = 8r^3 + 12r^2 + 6r + 1 - 8r^3 + 12r^2 - 6r + 1$$
$$= 24r^2 + 2 \text{ as req.}$$

23 b)

$$\sum_{r=1}^{n} 24r^2 + 2 = \sum_{r=1}^{n}((2r+1)^3 - (2r-1)^3) =$$

~~3^3~~	-1^3
~~$+5^3$~~	~~-3^3~~
~~$+7^3$~~	~~-5^3~~
~~$\ldots$~~	~~$\ldots$~~
~~$+(2n-3)^3$~~	~~$-(2n-5)^3$~~
~~$+(2n-1)^3$~~	~~$-(2n-3)^3$~~
$+(2n+1)^3$	~~$-(2n-1)^3$~~

$$= (2n+1)^3 - 1$$
$$= 8n^3 + 12n^2 + 6n + 1 - 1$$
$$= 8n^3 + 12n^2 + 6n$$
$$\therefore \sum_{r=1}^{n} r^2 = \frac{1}{24}\sum_{r=1}^{n}(24r^2+2) - \frac{1}{24}\sum_{r=1}^{n} 2$$
$$= \frac{1}{24}(8n^3 + 12n^2 + 6n) - \frac{1}{24}(2n)$$
$$= \frac{1}{24}n(8n^2 + 12n + 6 - 2)$$
$$= \frac{1}{6}n(2n^2 + 3n + 1)$$
$$= \frac{1}{6}n(2n+1)(n+1) \text{ as req.}$$

24 a)

$$(2r+1)^4 - (2r-1)^4 = 16r^4 + 32r^3 + 24r^2 + 8r + 1$$
$$- 16r^4 + 32r^3 - 24r^2 + 8r - 1$$
$$= 64r^3 + 16r \text{ as req.}$$

24 b)

$$\sum_{r=1}^{n} 64r^3 + 16r = \sum_{r=1}^{n}((2r+1)^4 - (2r-1)^4) =$$

~~3^4~~	-1^4
~~$+5^4$~~	~~-3^4~~
~~$+7^4$~~	~~-5^4~~
~~$\ddots$~~	~~$\ddots$~~
~~$+(2n-3)^4$~~	~~$-(2n-5)^4$~~
~~$+(2n-1)^4$~~	~~$-(2n-3)^4$~~
$+(2n+1)^4$	~~$-(2n-1)^4$~~

$$= (2n+1)^4 - 1$$
$$= 16n^4 + 32n^3 + 24n^2 + 8n + 1 - 1$$
$$= 16n^4 + 32n^3 + 24n^2 + 8n$$
$$\therefore \sum_{r=1}^{n} r^3 = \frac{1}{64}\sum_{r=1}^{n}(64r^3 + 16r) - \frac{1}{4}\sum_{r=1}^{n} r$$
$$= \frac{1}{64}(16n^4 + 32n^3 + 24n^2 + 8n) - \frac{1}{4}\left(\frac{1}{2}n(n+1)\right)$$
$$= \frac{1}{8}(2n^4 + 4n^3 + 3n^2 + n) - \frac{1}{8}n(n+1)$$
$$= \frac{1}{8}n(2n^3 + 4n^2 + 3n + 1 - n - 1)$$
$$= \frac{1}{8}n(2n^3 + 4n^2 + 2n)$$
$$= \frac{1}{4}n^2(n^2 + 2n + 1)$$
$$= \frac{1}{4}n^2(n+1)^2 \text{ as req.}$$

25 a)

$$(2r+1)^5 - (2r-1)^5$$
$$= 32r^5 + 80r^4 + 80r^3 + 40r^2 + 10r + 1$$
$$\quad - 32r^5 + 80r^4 - 80r^3 + 40r^2 - 10r + 1$$
$$= 160r^4 + 80r^2 + 2 \text{ as req.}$$

25 b)

$$\sum_{r=1}^{n} 160r^4 + 80r^2 + 2 = \sum_{r=1}^{n}((2r+1)^5 - (2r-1)^5) =$$

$$\begin{array}{ll}
\cancel{3^5} & -1^5 \\
\cancel{+5^5} & \cancel{-3^5} \\
\cancel{+7^5} & \cancel{-5^5} \\
\cancel{\ddots} & \cancel{\ddots} \\
\cancel{+(2n-3)^5} & \cancel{-(2n-5)^5} \\
\cancel{+(2n-1)^5} & \cancel{-(2n-3)^5} \\
+(2n+1)^5 & \cancel{-(2n-1)^5}
\end{array}$$

$$= (2n+1)^5 - 1$$
$$= 32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n + 1 - 1$$
$$= 32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n$$
$$\therefore \sum_{r=1}^{n} r^4 = \frac{1}{160}\sum_{r=1}^{n}(160r^4 + 80r^2 + 2) - \frac{1}{2}\sum_{r=1}^{n} r^2 - \frac{1}{80}\sum_{r=1}^{n} 1$$
$$= \frac{1}{160}(32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n)$$
$$-\frac{1}{2}\left(\frac{1}{6}n(n+1)(2n+1)\right) - \frac{1}{80}n$$
$$= \frac{1}{80}n(16n^4 + 40n^3 + 40n^2 + 20n + 5) - \frac{1}{12}n(n+1)(2n+1) - \frac{1}{80}n$$
$$= n\left[\frac{1}{5}n^4 + \frac{1}{2}n^3 + \frac{1}{2}n^2 + \frac{1}{4}n + \frac{1}{16} - \frac{1}{6}n^2 - \frac{1}{4}n - \frac{1}{12} - \frac{1}{80}\right]$$
$$= n\left[\frac{1}{5}n^4 + \frac{1}{2}n^3 + \frac{1}{3}n^2 - \frac{1}{30}\right]$$
$$= \frac{1}{30}n[6n^4 + 15n^3 + 10n^2 - 1]$$
$$= \frac{1}{30}n(n+1)[6n^3 + 9n^2 + n - 1]$$
$$= \frac{1}{30}n(n+1)(2n+1)(3n^2 + 3n - 1) \text{ as req.}$$

26)

$$\sum_{r=k}^{100}\left(\frac{2}{r} - \frac{2}{r+1}\right) =$$

$$\begin{array}{ll} \frac{2}{k} & -\cancel{\frac{2}{k+1}} \\ +\cancel{\frac{2}{k+1}} & -\cancel{\frac{2}{k+2}} \\ +\cancel{\frac{2}{k+2}} & -\cancel{\frac{2}{k+3}} \\ \cancel{\cdots} & \cancel{\cdots} \\ +\cancel{\frac{2}{98}} & -\cancel{\frac{2}{99}} \\ +\cancel{\frac{2}{99}} & -\cancel{\frac{2}{100}} \\ +\cancel{\frac{2}{100}} & -\frac{2}{101} \end{array}$$

$$= \frac{2}{k} - \frac{2}{101}$$
$$\therefore \frac{2}{k} - \frac{2}{101} = \frac{65}{1818}$$
$$\frac{2}{k} = \frac{1}{18}$$
$$\therefore k = 36$$

27 a)

$$\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2}$$
$$= \frac{2(r+1)(r+2)}{r(r+1)(r+2)} - \frac{r(r+2)}{r(r+1)(r+2)} - \frac{r(r+1)}{r(r+1)(r+2)}$$
$$= \frac{2r^2+6r+4}{r(r+1)(r+2)} - \frac{r^2+2r}{r(r+1)(r+2)} - \frac{r^2+r}{r(r+1)(r+2)}$$
$$= \frac{3r+4}{r(r+1)(r+2)} \text{ as req.}$$

27 b)

$$\sum_{r=1}^{n} \frac{3r+4}{r(r+1)(r+2)} = \sum_{r=1}^{n} \left(\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2} \right) =$$

$$\begin{array}{lll}
\frac{2}{1} & -\frac{1}{2} & -\cancel{\frac{1}{3}} \\
+\frac{2}{2} & -\cancel{\frac{1}{3}} & -\cancel{\frac{1}{4}} \\
+\cancel{\frac{2}{3}} & -\cancel{\frac{1}{4}} & -\cancel{\frac{1}{5}} \\
\cancel{\cdots} & \cancel{\cdots} & \cancel{\cdots} \\
+\cancel{\frac{2}{n-2}} & -\cancel{\frac{1}{n-1}} & -\cancel{\frac{1}{n}} \\
+\cancel{\frac{2}{n-1}} & -\cancel{\frac{1}{n}} & -\frac{1}{n+1} \\
+\cancel{\frac{2}{n}} & -\frac{1}{n+1} & -\frac{1}{n+2}
\end{array}$$

$$= 2 - \frac{1}{2} + \frac{2}{2} - \frac{1}{n+1} - \frac{1}{n+1} - \frac{1}{n+2}$$
$$= \frac{5}{2} - \frac{2}{n+1} - \frac{1}{n+2}$$

27 c)

$$\sum_{r=k}^{25} \frac{3r+4}{r(r+1)(r+2)} = \sum_{r=k}^{25} \left(\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2} \right) =$$

$$\begin{array}{lll}
\frac{2}{k} & -\frac{1}{k+1} & -\cancel{\frac{1}{k+2}} \\
+\frac{2}{k+1} & -\cancel{\frac{1}{k+2}} & -\cancel{\frac{1}{k+3}} \\
+\cancel{\frac{2}{k+2}} & -\cancel{\frac{1}{k+3}} & -\cancel{\frac{1}{k+4}} \\
\cancel{\cdots} & \cancel{\cdots} & \cancel{\cdots} \\
+\cancel{\frac{2}{23}} & -\cancel{\frac{1}{24}} & -\cancel{\frac{1}{25}} \\
+\cancel{\frac{2}{24}} & -\cancel{\frac{1}{25}} & -\frac{1}{26} \\
+\cancel{\frac{2}{25}} & -\frac{1}{26} & -\frac{1}{27}
\end{array}$$

$$= \frac{2}{k} - \frac{1}{k+1} + \frac{2}{k+1} - \frac{1}{26} - \frac{1}{26} - \frac{1}{27}$$
$$= \frac{2}{k} + \frac{1}{k+1} - \frac{40}{351}$$
$$\therefore \frac{2}{k} + \frac{1}{k+1} - \frac{40}{351} = \frac{1174}{12\,285}$$
$$\frac{2(k+1)}{k(k+1)} + \frac{k}{k(k+1)} = \frac{22}{105}$$
$$2k + 2 + k = \frac{22}{105}k(k+1)$$
$$315k + 210 = 22k^2 + 22k$$
$$22k^2 - 293k - 210 = 0$$

which has solutions $k = -\frac{15}{22}$ and $k = 14$

$$\therefore k = 14$$

28 a)

$$2^{r+1}(r-1) - 2^r(r-2) = 2 \times 2^r(r-1) - 2^r(r-2)$$
$$= 2 \times 2^r \times r - 2 \times 2^r - 2^r \times r + 2 \times 2^r$$
$$= 2 \times 2^r \times r - 2^r \times r$$
$$= r2^r \text{ as req.}$$

28 b)

$$\sum_{r=1}^{n} r2^r = \sum_{r=1}^{n}(2^{r+1}(r-1) - 2^r(r-2)) =$$

~~$2^2 \times 0$~~	$-2^1 \times -1$
~~$+2^3 \times 1$~~	~~$-2^2 \times 0$~~
~~$+2^4 \times 2$~~	~~$-2^3 \times 1$~~
~~$\therefore$~~	~~$\therefore$~~
~~$+2^{n-1}(n-3)$~~	~~$-2^{n-2}(n-4)$~~
~~$+2^n(n-2)$~~	~~$-2^{n-1}(n-3)$~~
$+2^{n+1}(n-1)$	~~$-2^n(n-2)$~~

$$= 2 + 2^{n+1}(n-1) \text{ as req.}$$

29 a)

$$\ln\left(1 + \frac{1}{r+1}\right) = \ln\left(\frac{r+1}{r+1} + \frac{1}{r+1}\right)$$
$$= \ln\left(\frac{r+2}{r+1}\right) = \ln(r+2) - \ln(r+1)$$

29 b)

$$\sum_{r=1}^{n} \ln\left(1+\frac{1}{r+1}\right) = \sum_{r=1}^{n} (\ln(r+2) - \ln(r+1)) =$$

$$\begin{array}{rr} \cancel{\ln 3} & -\ln 2 \\ \cancel{+\ln 4} & \cancel{-\ln 3} \\ \cancel{+\ln 5} & \cancel{-\ln 4} \\ \cancel{+\ldots} & \cancel{-\ldots} \\ \cancel{+\ln(n)} & \cancel{-\ln(n-1)} \\ \cancel{+\ln(n+1)} & \cancel{-\ln n} \\ +\ln(n+2) & \cancel{-\ln(n+1)} \end{array}$$

$$= \ln(n+2) - \ln 2$$

$$= \ln\left(\frac{n+2}{2}\right)$$

30)

$$\sum_{r=1}^{n} \ln\left(\frac{r+1}{r}\right) = \sum_{r=1}^{n} (\ln(r+1) - \ln(r)) =$$

$$\begin{array}{rr} \cancel{\ln 2} & -\ln 1 \\ \cancel{+\ln 3} & \cancel{-\ln 2} \\ \cancel{+\ln 4} & \cancel{-\ln 3} \\ \cancel{+\ldots} & \cancel{-\ldots} \\ \cancel{+\ln(n-1)} & \cancel{-\ln(n-2)} \\ \cancel{+\ln(n)} & \cancel{-\ln(n-1)} \\ +\ln(n+1) & \cancel{-\ln n} \end{array}$$

$$= \ln(n+1) - \ln 1 = \ln(n+1)$$

31)

$$\sum_{r=21}^{50} \ln\left(\frac{r+1}{r-1}\right) = \sum_{r=21}^{50} (\ln(r+1) - \ln(r-1)) =$$

$$\begin{aligned} &\cancel{\ln 22} && -\ln 20 \\ &\cancel{+\ln 23} && -\ln 21 \\ &\cancel{+\ln 24} && \cancel{-\ln 22} \\ &\cancel{+\ldots} && \cancel{-\ldots} \\ &\cancel{+\ln 49} && \cancel{-\ln 47} \\ &+\ln 50 && \cancel{-\ln 48} \\ &+\ln 51 && \cancel{-\ln 49} \end{aligned}$$

$$= \ln 50 + \ln 51 - \ln 20 - \ln 21 = \ln\left(\frac{85}{14}\right)$$

32 a)

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right) = \sum_{r=1}^{n} (\ln(r+3) - \ln(r+1)) =$$

$$\begin{aligned} &\cancel{\ln 4} && -\ln 2 \\ &\cancel{+\ln 5} && -\ln 3 \\ &\cancel{+\ln 6} && \cancel{-\ln 4} \\ &\cancel{+\ldots} && \cancel{-\ldots} \\ &\cancel{+\ln(n+1)} && \cancel{-\ln(n-1)} \\ &+\ln(n+2) && \cancel{-\ln n} \\ &+\ln(n+3) && \cancel{-\ln(n+1)} \end{aligned}$$

$$\begin{aligned} &= \ln(n+2) + \ln(n+3) - \ln 2 - \ln 3 \\ &= \ln((n+2)(n+3)) - \ln 6 \\ &= \ln\left(\frac{1}{6}(n+2)(n+3)\right) \end{aligned}$$

32 b)

$$\begin{aligned} \sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right)^{36} &= \sum_{r=1}^{n} 36 \ln\left(\frac{r+3}{r+1}\right) \\ &= 36 \ln\left(\frac{1}{6}(n+2)(n+3)\right) \end{aligned}$$

33 a)

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) = \sum_{r=1}^{n} (\ln(2r+3) - \ln(2r+1)) =$$

$$\begin{array}{ll} \cancel{\ln 5} & -\ln 3 \\ \cancel{+\ln 7} & \cancel{-\ln 5} \\ \cancel{+\ln 9} & \cancel{-\ln 7} \\ \cancel{+\cdots} & \cancel{-\cdots} \\ \cancel{+\ln(2n-1)} & \cancel{-\ln(2n-3)} \\ \cancel{+\ln(2n+1)} & \cancel{-\ln(2n-1)} \\ +\ln(2n+3) & \cancel{-\ln(2n+1)} \end{array}$$

$$= \ln(2n+3) - \ln 3$$

$$= \ln\left(\frac{1}{3}(2n+3)\right)$$

33 b)

$$\ln\left(\frac{1}{3}(2n+3)\right) > 4$$

$$\frac{1}{3}(2n+3) > e^4$$

$$2n+3 > 3e^4$$

$$n > \frac{1}{2}(3e^4 - 3) \therefore n = 81$$

34)

$$\sum_{r=6}^{n}\left(\frac{1}{r+1} - \frac{1}{r+3}\right) =$$

$$\begin{array}{ll} \frac{1}{7} & \cancel{-\frac{1}{9}} \\ +\frac{1}{8} & \cancel{-\frac{1}{10}} \\ \cancel{+\frac{1}{9}} & \cancel{-\frac{1}{11}} \\ \cancel{\cdots} & \cancel{\cdots} \\ \cancel{+\frac{1}{n-1}} & \cancel{-\frac{1}{n+1}} \\ \cancel{+\frac{1}{n}} & -\frac{1}{n+2} \\ \cancel{+\frac{1}{n+1}} & -\frac{1}{n+3} \end{array}$$

$$= \frac{1}{7} + \frac{1}{8} - \frac{1}{n+2} - \frac{1}{n+3}$$

As $n \to \infty$, $\frac{1}{n+2} \to 0$, $\frac{1}{n+3} \to 0$

$$\therefore \sum_{r=6}^{\infty}\left(\frac{1}{r+1} - \frac{1}{r+3}\right) = \frac{1}{7} + \frac{1}{8} = \frac{15}{56}$$

35 a)

$$\frac{1}{2(r+2)} - \frac{1}{2(r+4)} = \frac{r+4}{2(r+2)(r+4)} - \frac{r+2}{2(r+2)(r+4)}$$
$$= \frac{2}{2(r+2)(r+4)} = \frac{1}{(r+2)(r+4)} \text{ as req.}$$

35 b)

$$\sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} = \sum_{r=1}^{n} \left(\frac{1}{2(r+2)} - \frac{1}{2(r+4)} \right) =$$

$$\begin{array}{ll} \frac{1}{2\times 3} & -\cancel{\frac{1}{2\times 5}} \\ +\frac{1}{2\times 4} & -\cancel{\frac{1}{2\times 6}} \\ \cancel{+\frac{1}{2\times 5}} & -\cancel{\frac{1}{2\times 7}} \\ \cancel{\cdots} & \cancel{\cdots} \\ \cancel{+\frac{1}{2n}} & \cancel{-\frac{1}{2(n+2)}} \\ \cancel{+\frac{1}{2(n+1)}} & -\frac{1}{2(n+3)} \\ \cancel{+\frac{1}{2(n+2)}} & -\frac{1}{2(n+4)} \end{array}$$

$$= \frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)}$$
$$\frac{7}{24} - \left(\frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)} \right) < \frac{19}{180}$$
$$\frac{1}{2(n+3)} + \frac{1}{2(n+4)} < \frac{19}{180}$$
$$\frac{n+4}{2(n+3)(n+4)} + \frac{n+3}{2(n+3)(n+4)} < \frac{19}{180}$$
$$\frac{2n+7}{(n+3)(n+4)} < \frac{19}{90}$$
$$180n + 630 < 19(n+3)(n+4)$$
$$180n + 630 < 19n^2 + 133n + 228$$
$$19n^2 - 47n - 402 > 0$$
$$n < -\frac{67}{19} \text{ or } n > 6$$

n is a positive integer number so $n \geq 7$ as req.